Xiao Zhou

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EDUCATION

Shanghai Jiao Tong University
B.S./BEng. in Artificial Intelligence — Sep. 2023 – Expected Jun.2027
GPA: 91.6/100 **Rank: 9/98**
Main courses: Linear Algebra(99), Mathematical Analysis I(92), Mathematical Analysis II(97), Design and Analysis of Algorithms(98), Discrete Mathematics(92), Probability and Statistics(93), Linear and Convex Optimization(93), Deep Learning(90), Stochastic Processes(95), Computer Architecture(94), Computer Networks(94), Reinforcement Learning(91), Mathematical Logic(96), Large Language Models(96), Natural Language Processing(99)

PUBLICATION

X. Zhou*, S. Zhang*, et al., "GUI vs. CLI: Execution Bottlenecks in Screen-Only and Skill-Mediated Computer-Use Agents", *Under review at EMNLP 2026*, 2025.

S. Zhang*, Y. Gao*, **X. Zhou***, et al., "MRMR: A Realistic and Expert-Level Multidisciplinary Benchmark for Reasoning-Intensive Multimodal Retrieval", in *Proceedings of the International Conference on Learning Representations (ICLR 2026)*, 2025.

J. Wei, Q. Ma, Y. Zhao, **X. Zhou**, et al., "OpenComputer: Verifiable Software Worlds for Computer-Use Agents", in *Under review at NeurIPS 2026*, 2026.

RESEARCH EXPERIENCE

Research Focus: Agent, AI4Research — Mar. 2026 – Present
Research Assistant | *Advisor:* Prof. Arman Cohan, Yale University

- Built a benchmark for forecasting research weak signals and evaluated diverse research systems on their ability to identify and anticipate emerging research topics.

Research Focus: Multimodal Reasoning, Agent — Jun. 2025 – Present
Research Assistant | *Advisor:* Prof. Chen Zhao, New York University Shanghai

- Constructed a matched execution-layer benchmark with curated datasets to compare screen-only and skill-mediated computer-use agents under identical goals and states, revealing modality-specific execution bottlenecks and differences in full-task success.
- Contributed to a research project that developed a realistic, expert-level multidisciplinary benchmark for reasoning-intensive multimodal retrieval, involving the design of complex query–corpus structures, and LLM-based reasoning baselines to better measure models' real-world retrieval capabilities.
- Investigated image-based memory representations for agents, exploring different compression strategies and layout designs for encoding interaction history, and evaluating their impact on task performance and execution efficiency.

Research Focus: Large Language Models Reasoning, Efficiency Mar. 2025 – Present
Research Assistant | *Advisor:* Prof. Hai Zhao, Shanghai Jiao Tong University, School of Computer Science

- Contributed to a project systematically analyzing the impact of efficiency techniques, including quantization, pruning, sparsification, and KV-cache compression, on the safety, robustness, and reasoning capabilities of Large Reasoning Models. Designed controlled experiments and evaluated trade-offs between computational efficiency and model performance.
- Explored a project exploring the super weights in LLMs that underpin mathematical reasoning, using techniques such as weight attribution analysis, neuron activation tracing, and targeted ablation to uncover key parameter groups responsible for mathematical reasoning ability.

Research Focus: LLMs, AI4Chemistry (AI4Chem) Nov. 2024 – Mar. 2025
Research Assistant | *Advisor:* Prof. Junchi Yan, Shanghai Jiao Tong University, School of Artificial Intelligence

- Manually implemented a Transformer model, reproducing key architectural components such as multi-head self-attention, positional encoding, and training optimization mechanisms.
- Conducted multiple data analysis and model training tasks, including dataset preprocessing, exploratory data analysis, and evaluating model performance.

INDUSTRY EXPERIENCE

Yalotein Biotech Sep. 2025 – Present
Intern

- Developed a knowledge-base management and question-answering application with document ingestion, keyword and embedding-based retrieval, user-driven knowledge updates, and an interactive querying interface.
- Developing a multi-agent system with a focus on the knowledge-base agent.

SKILLS

Programming Languages: Python, C++
Languages: Chinese (native), English (proficient)